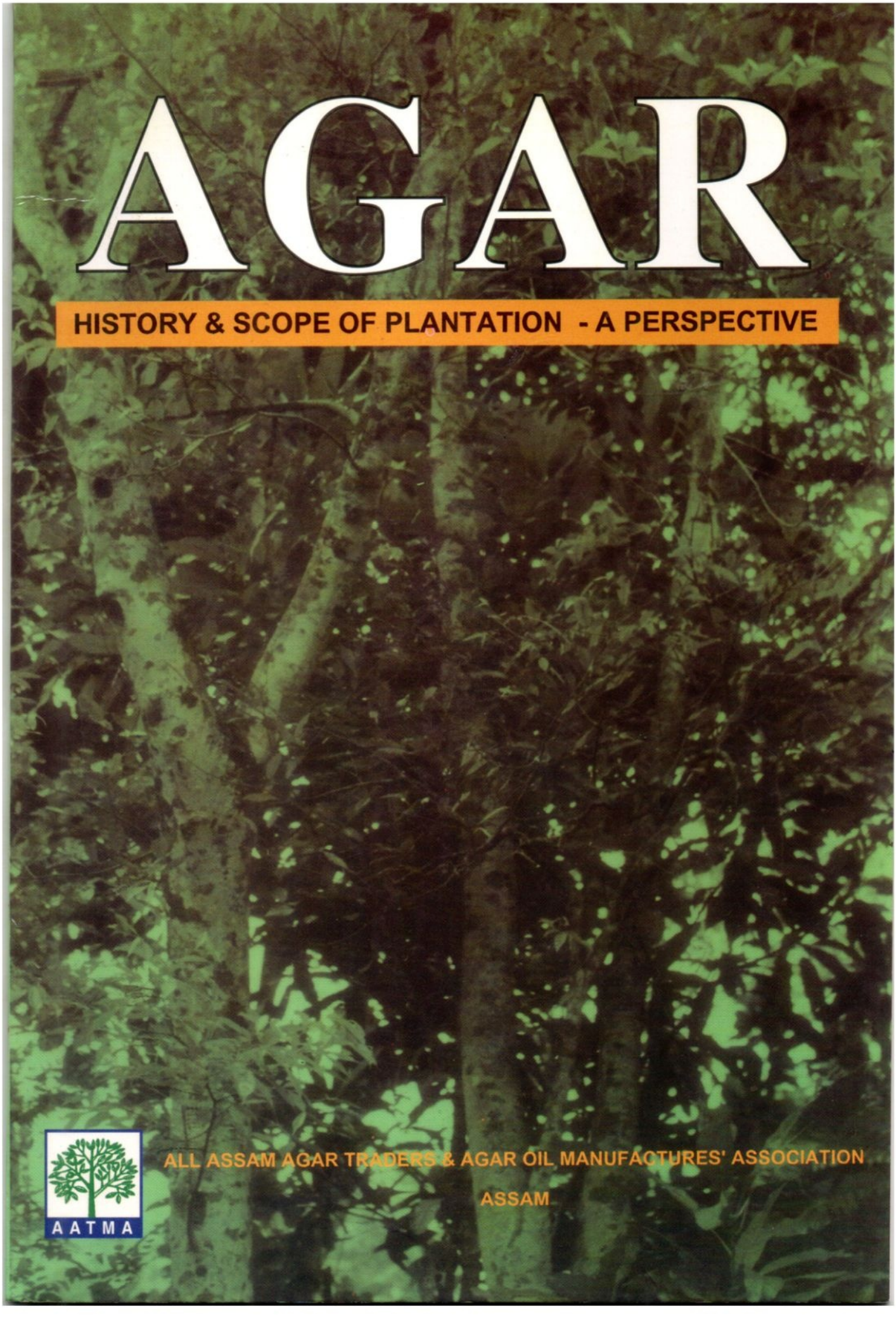

AGAR
HISTORY & SCOPE OF PLANTATION - A PERSPECTIVE
AATMA
ALL ASSAM AGAR TRADERS & AGAR OIL MANUFACTURES' ASSOCIATION
ASSAM

# AGAR

## HISTORY & SCOPE OF PLANTATION
## A PERSPECTIVE

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**All Assam Agar Traders & Agaroil Manufacturers' Association**
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Sanchi plants in the compound of Sri Sri Batadrava Than,
(Bar-Hissha) Batadrava, Nagaon. Assam.

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# A FEW WORDS FROM THE PUBLISHER

Agar Trading and Agar oil Extraction in Assam surfaced as a totally unorganized and rural based cottage industries run by the family members. With the expansion in this venture attempts were made to organize and improve upon through various moves like Co-operative approach, forming Companies etc. In an attempt to bring the venture under one platform spread all around All Assam Agar Traders and Agar Oil Manufactures' Association came into being in the year 1976 with its HO at Hojai, where the highest concentration of Agar distillation units in the Country exists. Since then the Association engaged itself for various development activities of the venture. Presently, the Association has taken up elaborate awareness campaign for large scale commercial plantation, conservation of the species in their natural habitat (wild) and replacing the existing distillation units by fuel efficient units under the technical guidance of the R&D centre of Ajmal Group at Hojai.

In view of the present crisis on the subject, particularly on the reduction of agar population in wild at an alarming rate, enlisting the species in the Appendix II to the Convension on International Trade in Endangered Species of fauna & flora (CITES) and declaration of *Aquilaria malaccensis* by the Govt. of India as endangered species, and subsequently the partial restrictions imposed upon, the Association submitted a memorandum to the Hon'ble Chief Minister of Assam on 26th June/2000 with the following prayers: (a) to introduce and declare Agar as a plantation crop and the trading and oil extraction as agro-based industry for the greater interest of the state. In the memorandum we enclosed a "Brief note on "Agarwood (*Aquilaria spp.*) and its Scope as a Plantation Crop in Assam" prepared by Dr. M. Ahmed and Dr. P. Gogoi. The Association later decided to publish the same in the form of a booklet, which may be of interest to our members and other users.

While we appreciate the steps taken up by the Govt. and various agencies to conserve this valuable species, at the same time with our years long experience on the species and its present status we cannot agree that Agar tree is an endangered species. Rather we feel that the economic exploitation on scientific basis can only save and develop the species. With the shortage of raw material supply already it could attract many to go for commercial

cultivation in the NE region. It has come to our knowledge that the authors have prepared a practical manual "Agar plantation in Assam-Package of Practices" useful for growers/planters incorporating all the necessary plantation technologies. I congratulate them for this noble job at right time.

In view of the above, we would draw the kind attention of our Govt. and all concerned to a proposal of forming a corporation "Assam Agar Plantation Corporation" a separate agency under the Govt. patronization for its conservation in wild and overall development for sustainable economy of the state. Agar could successfully put Assam in the international scenario. Therefore, its conservation and development by all means are necessary to keep it up.

We offer our heartfelt thanks to the authors Dr. M. Ahmed and Dr. P. Gogoi for compiling various valuable informations on the subject generated out of extensive studies and research carried out at the centre since 1995. We are also thankful to Mr. N.K.Vasu, D.F.O, Nagaon South Division, Hojai for his keen interest and co-ordinating with the R&D section for artificial regeneration of the species with special reference to seed technology.

I hope this booklet will provide the required impetus to the hitherto unattended aspects on Agar and furnish some valuable information to those involved in Plantation, Industry and Trade.

Date : 25th September, 2000
Place : Hojai

**Ataur Rahman**
President
All Assam Agar Traders and Agaroil
Manufacturers' Association,
Hojai, Dist. Nagaon, Assam-782 435

# INTRODUCTION

Sanchi plant or the Agarwood tree is one of the precious gift of nature deeply associated with the cultural heritage of Assam since ancient time. During early period the specially treated bark of agar tree popularly called 'Sanchipat' was used for writing religious scripts which are still being preserved in many places including different 'Satra/Namghar' in Assam.

It belongs to the genus *Aquilaria* of family Thymeleaceae. The genus includes the economically exploited species *Aquilaria agallocha*, *A. malaccensis* and *A. khasiana* in the North East India of which *A. agallocha* found the most exploited one. The agarwood (black resinous wood) or 'agaru' and agar oil (the essential oil) or agar attar are the most exalted perfumery raw materials obtained from the infected wood of this *Aquilaria spp.* The agar wood oil or aloe wood oil, known in the east as Agar attar is obtained by distilling selected parts of the infected wood of *Aquilaria spp.* The oil is one of the perfumery's oldest materials used in high-class perfumery and as a fixative, imparting a lasting balsamic odour to the product. The agarwood or 'agaru' on burning emits sweet fragrance that has no parallel in the world.

# HISTORY OF AGARWOOD

The use of 'agaru' is pre-historic. The Aloe wood of the Bible was the heartwood of *Aquilaria ovata* and *A. agallocha.* The Aloes referred to in the Bible was evidently a very aromatic plant and most probably the agarwood. There are mentions on the use of Aloe wood (Udul-Hind) in Paradise as incense in the famous Ahadith-Sahi Al-Bukhari. There are mentions of 'agaru' of ancient

Kamrup in the report of Chinese Pilgrim Hiuen-Tsang, Abhijnanam Sakuntalam of Kalidas, and Arthosasthra of Koutilya. The first historical biographies in Sanskrit the Harshacharita written by Bana in 652 AD states that the presents sent by Bhaskaravarman to Harsha included among other things, volumes of fine writings in leaves made from aloe bark ( bark of agar plant) and black aloe oil. There is mention in the Sabhaparvan of the Mahabharata that in the course of Digvijaya Bhimsena went to Pragjyotish and received sandalwood and aloewood (agaru) as presents. The Nowgong grant of Balavarman gives a graphic description of Pragjyotishanagara where arecanut trees were wrapped in leaves of creeper of betel-plants and Krishnaguru (Telegu or Tamil name of Agarwood) or black aloe-wood trees were surrounded with cardamom creepers.

After conquering the capital of last king Gaur Govind in 1384 A.D in Sylhet Saint Fakir Shah Jalal (RA) and his followers found agar wood and agar attar along with many other valuables in the Royal store. This indicates that distillation of agar oil was done during thirteen century or even early in India. Abul Fazal Allami in his Ain-I-Akbari (memoir of Emperor Akbar) written in about 1590 AD. gave a vivid description of agarwood and agar oil along with their manufacturing process and uses. It is also said that Mughal invaded Assam mainly for 'agaru'.

From Kamrup 'agaru' had been exported to the Middle East from time immemorial may be by the Chinese traders through the Silk road which extended from China to Middle-East through Kamrup and then India. During those days in Kamrup 'Agaru' and Chandan (*Santalum album* L.) were the main items of cosmetics as there are mentions of these two articles in different old scriptures of Kamrup like Ramayan translated by Madhava Kandali. The 15th century Saint-Reformer and Literary Giant Sri Sankardev used bark of the tree as 'Sanchipat' for writing religious scripts which is still being preserved in many places. In a devotional verse, he described the 'Agaru' and the Chandan plant as divine, capable of fulfilling human desires. In folk songs also there are mentions of 'agaru'.

During 1900 plentiful extraction of the perfumed wood (agaru)

was done in various parts of undivided Assam. Assam 'agaru' used to go to Calcutta and from there to Turkey, Arabia, Parsia and Europe. At present Indian 'agaru' is largely exported to Arab countries where it is used as incense and also in the manufacture of joss sticks.

## COMMERCIAL DEVELOPMENT

### PRE-INDEPENDENCE PERIOD

The exploitation of agarwood was a customary in undivided Assam as early as 1905-1906. At that time the inhabitant of villages Dakhshinbagh, Suja Nagar and Rafi Nagar in Sylhet (now in Bangladesh), had the monopoly of the trade in Agarwood and agar oil. Majority of the distilleries was in the village Suja Nagar. Commercial distillation and trading on Agarwood oil in this sub-continent thus, started in Sylhet in the early twentieth century. Thus North East India, the major sources of raw material, provided an important cottage industry in the Sylhet region. During those days 'Agar Mahals' were leased out to the agarwood collectors in a defined divisions and for a fixed period through the Forest Department.

## POST- INDEPENDENCE PERIOD

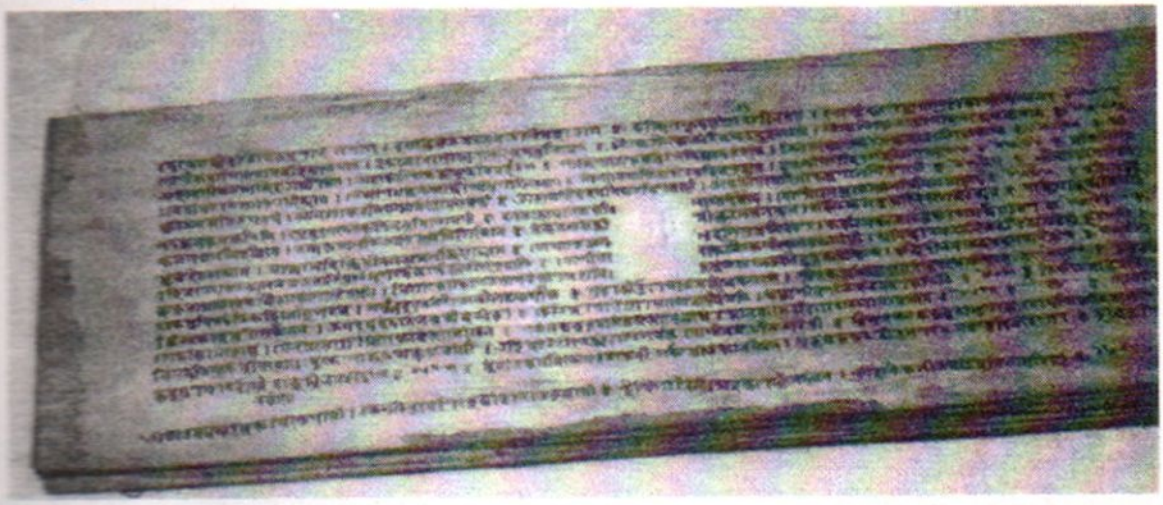
After partition the agar business in Sylhet came to a halt. Most of the agar growing forests occurred in India while the distilleries were in East Pakistan which later liberated as Bangladesh. Then some of the leading business families who were already in the business in Hojai and in other places of Assam since 1940 expanded their business by setting up distillation units on commercial scale in and around Hojai and other parts of upper Assam following this sudden change. Since then, agar distillation started on commercial venture mostly as rural base cottage industry in this part of the country. It is the endeavour of Haji Ajmal Ali, Haji Anfar Ali, Haji Mustafa Uddin and some other families whose relentless efforts could elevate the venture to its present status in India and abroad.

At that time, Forest Department used to lease out 'Agar Mahals' in clearly defined divisions in many states in the NE region viz, Naga Pahar, Mizo Pahar, Khasi Pahar, NEFA (Arunachal Pradesh), Garo hills, Mikir hills (Karbi Anglong) etc. Bhutan Govt. also continued leasing of 'Agar Mahals' till 1976. The 'Agar Mahals' were leased out for 2-3 years period for collection of agarwood only. This 'Mahal' system continued till 1975-76 after which it was discontinued in most of the states. Mizoram continued leasing till 1993 last. Since then the industry is running on the raw material supply mostly from private holdings and import.

After 1975 in search of alternative sources people started planting of agarwood tree in private holdings. Some of the big houses started importing agarwood from Singapore since 1982. Singapore is considered as the centre for importing agarwood of Indonesian and Malaysian origin. At present the major supply of raw material in Hojai is made available through import. Only the small distilleries are depending on local supply mainly from private holdings. By now, a large number of units in the form of cottage industries as well as in S.S.I unit levels are engaged in this business. Although Hojai in Nagaon district is regarded as the main agar business centre in the country the other places where it took the shape of cottage industries are- Naharani in Golaghat district, Namti , Charaideu, Amguri of Sibsagar and Kakojan of Jorhat district of upper Assam. Sri Monmohan Dev and Sri Jugesh Dev of Barpather, Sri Akhil Barman of Dhansiri and Sri Abdul Shahid in Naharani of Golaghat district were the pioneers in this venture started as early as 1940 in those areas. The first distillation unit in upper Assam (Copper deg) was set up in Sibsagar District comprising the then Golaghat Sub-division in the Forties. At present in Sibsagar most of the distillation units are run by natural gas supplied through underground pipeline which makes oil extraction comparatively cheaper. The trade in agar remained to a large extent an eastern trade and currently can be geographically divided into trade in Indian sub-continent, the Middle East and in South East Asia.

## USE OF SANCHIPAT

Agar tree in Assam is known as Sanchi or Agar plant. During ancient time the bark was a source of writing-material known as Sanchipat which was used for writing history, mythology, folk songs, on medicines and holy scripts. The bark after giving a special treatment becomes immune to attack by insects and other microbes. For making long lasting, these were treated with arsenic sulphate an ore found in Himalayan region commonly known as Haintal in Assamese. Because of the bark use as 'Sanchipat' agar plant was named as Sanchi plant. For writing on these bark an indelible indigenous herbal ink were in use.



255 years old 'Sanchiputhi' written on Sanchipat.
Courtegy: Sri Sri Narowa Balisatra Parichalana Samiti,
P.O. Balisatra, Dist. Nagaon, Assam.

The ink was prepared as follows: Fruits of *Emblica officinalis* Gaertn. (Amla) and *Terminalia chebula* Retz. (Haritaki) and aerial parts of *Eclipta alba* (L) Hassk. (Kehraj) in equal proportion were crushed and extracted with water. Iron in the ferric form was added to this extract. The solution (the ink) was prepared in earthen pots.

## DISTRIBUTION

The genus *Aquilaria* of family Thymeleaceae comprises 8 species, which are occurring in Eastern Himalayas upto an altitude of 2000 m, China, South East Asia. Three species viz., *A. agallocha, A. malaccensis* and *A. khasiana* are found in NE India. Among

them *A. agallocha* Roxb. is the most dominant and economically exploited one. It occurs in parts of Bangladesh, Bhutan , and along the foot hills and sub-hilly areas in ever green and semi-evergreen forests in most of the states of North East India, including West Bengal and Sikkim. In the NE states agar tree is found to be distributed in Assam, Arunachal Pradesh, Nagaland, Mizoram, Manipur, Tripura and Meghalaya in wild as well as in sporadic cultivation in private sector. It is more predominantly distributed in the areas bordering Myanmar. The tree is found in certain parts of China. It is valued mainly for the brown to black oleoresin deposits that are found in about 10 –15 % of the naturally occurring tree of *A. agallocha* and other species of the genus *Aquilaria.* In Assam the concentration of natural distribution is found in Sibsagar, Golaghat, Dibrugarh and North Lakhimpur followed by Karbi Anglong and N.C.Hills. Agar plant is also found in Bhutan adjoining areas now under Kokrajhar, Nalbari, Mangaldoi and Sunitpur districts.

## THE TREE

It is a large to medium evergreen tree 15-20 m high, sometimes grow upto 40 m in height as is found in Barak Valley, 1.5-2.5 m girth with a moderately straight and often fluted stem. Leaves 5-9 cm long, thinly cariaceous, oblong lanceolate; flowers white or green to dirty yellow in terminal, sessile or shortly peduncled umbellate cymes. The tree regenerates freely by seeds. The fungus infected trees furnish the agarwood or eaglewood of commerce which occurs as dark coloured resinous fragrant masses in the centre of the bole and branches. The normal (uninfected) wood is soft, light and elastic. It is white to pale yellowsh white and has no particular odour.

## HOW THE TREE BECOMES SO VALUABLE

The tree becomes valuable only after getting infected by a particular fungus or group of fungi, ceases to grow and become sick in the population stand. The agar oil or 'Agaru' is thus a product

of disease caused by certain fungus. The infection occurs when stem is injured or bored by larvae of a particular stem borer (*Zeuzera conferta* Walker) belonging to the family Lepidoptera. These borers make vertical tunnels (hollow and zigzag) inside the tree trunk and thus the surfaces of the tunnels become the initial sites for infections. Later on infections spread on all sides slowly and gradually and ultimately a larger wood volume become infected.



The young larva (tree borer) that makes tunnels inside the tree and starts the process.

Infection may also occur due to mechanical or natural injuries on the stem or branches. Due to infections oleoresins are accumulated in the infected wood and later become odoriferous. At the initial stage infection appears as brown streaks in the tissue. Accumulation of oleoresin goes on increasing with the increase of infection area, as well as aging of infection. More oleoresin deposits which results increase in the depth of colour of infected wood and finally it become brown to black. Heavy and old age infection may lead to death of the plant. Thus, the yield of 'agaru' depends on insect–fungal interaction on the host plant since there is no special cells or glands to synthesize the oil, as found in other essential oil bearing plants.

The specific fungi being soil borne builds up population with the increase in plantation.



Natural injury- a way of agar oil/ agaru formation

## RAISING PLANTATION

The non-availability of infected trees in the natural population and increasing demand and footing of Cottage to SS industries in many places are some factors attracting many people to go for private plantation in the NE region. Nagaland and Tripura Govt. have undertaken plantation programmes and also encouraging private planting of agar. In Assam many commercial plantations are coming up every year.

## SOIL AND CLIMATE

Climate decides eco-system of a region. Maximum manifestation of a particular plant species is attained only under

optimum climatic conditions. Agar plant (Sanchi plant) prefers warm and high humid sub-tropical climate with rainfall ranging from 1800-3500mm per annum. It grows from sea level to 2000-m altitude. It grows on well-drained deep sandy loam to loam rich in organic matter and in marginal soils and also in shallow soils over rocky beds. It grows well on hill slopes and forest environment. The traditional agar growing areas show that it prefers acidic soil reaction. The mycorrhiza and other beneficial fungi which seems to be responsible for normal growth and oil synthesis in agar tree being soil borne requires acid soil for their population build up. Agar tree does not thrive at all in water- logged condition. The primary factors for growth are internal drainage and soil depth.

Sanchi plant is sun loving and rarely found in deep forests. It produces good quality oil and agarwood when grown on the south slope of a hill/hillock receiving high amount of sunshine. They grow on the borders of forests, scrub jungles and in homesteads.

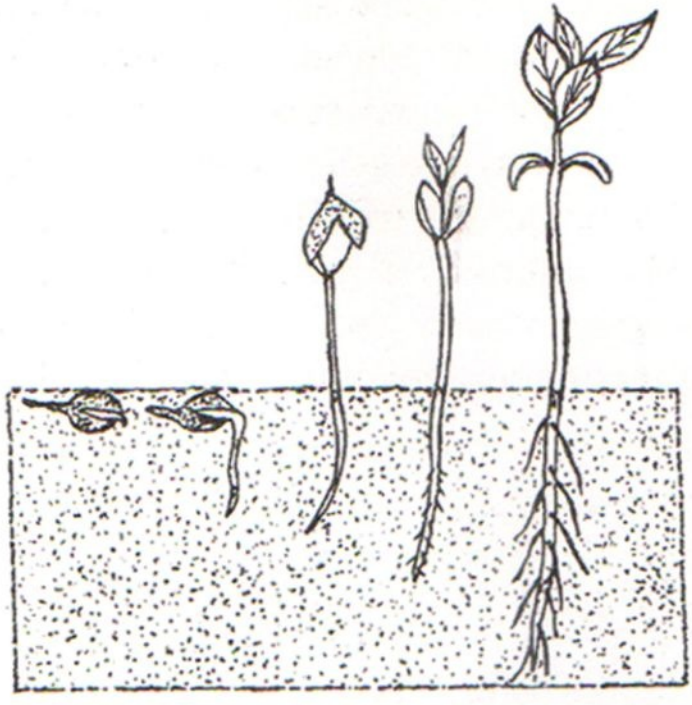
## PROPAGATION

Agar is propagated by seeds. It is self-pollinated species, and about 7-10% cross-pollination may occur. Agar tree flowers in April-May and fruiting begins from June-July. The mature seeds become available during July-August. Because the mature seeds become available during peak monsoon, natural regeneration is not a problem. The fresh seeds can be stored for 5-6 days in moist sand and more than two months in cold storage. Care should be taken so that seeds donot get dried during this period. Dry seeds loose their viability rapidly. The seeds remain viable for short period of 10-15 days under ordinary condition.

## SEED GERMINATION

The germination is epigeal and starts from 7-10th day and may continue upto 30-35 days in beds. In the field condition the per cent germination is about 40-50% while under ideal situation in sand beds it is above 85-90 % with one to four weeks dormancy. A

maximum dormancy period of 5 weeks has also been recorded. For commercial plantation seeds are first germinated in beds and at 25 days onward seedlings are transferred to polybags where it is nursed for about one to two years till they are planted in the main field .



Stages of seedling emergence

## PLANTING

After land preparation 1-2 years old seedling are planted in the main field at about 2.5-3.0 m spacing either side depending upon



Year old seedlings raised in poly bags ready for planting

situation during monsoon (May-Sept). In absence of rain watering has to be done immediately after planting. Staking should be done to keep the seedlings intact and upright.

## SILVICULTURAL CHARACTERISTICS

Agar tree is a light demander though it can stand shade from a high cover during early stages of growth. In hilly areas planting on the south side gives best results.

Agar tree responses very well to coppicing and pollarding when they are young and attain a certain growth (after 4-5 years onwards). This characteristics facilitates harvesting of infected tree leaving the crown or tree trunk for quick regeneration for a second crop (ratoon crop) and seed production from the coppiced/pollarded tree once identified as a good mother plant from quality and production point of view.



A coppiced shoot growing for a second harvest.

***Qushte Bahari (White aloewood) is the best among those used in medicines"***

*—Bukhari & Mishkat Sharif*

Coppicing is done during March-May for maximum regeneration. The coppiced tree grows faster because of already established root system that covers a larger soil volume for tapping water and nutrients. This tree gets infected early with a higher yield of oil extractable chips.

## AGAR IN HOMESTEAD PLANTING

In upper Assam one can see particularly in rural areas that the front side of a residential area has been demarcated by growing agar trees. In many houses agar has been growing as avenue tree and also planting in available space around homestead areas. Agar trees are sometimes grown for their economic importance or aesthetic or both. The value of agar trees as attractive to a house has become increasingly evident. In homestead area tree grows luxuriantly utilizing the waste and drained water around. Secondly, after attaining at harvestable stage security has been a big concern now a day particularly for small/marginal farmers. Therefore, farmers prefer to grow around available homestead areas, Kathonibari, garden boundaries etc. rather than raising an isolated plantations.



A 20 years old Sanchi plant in the compound of Sri Sri Batadrava Than (Bar-Hissha), Batadrava (Nagaon) Assam)

## AGAR IN AGRO-FORESTRY

Agar tree is suitable for growing on field boundaries and for dividing whole plot into sub-plots. Not only this, agar tree is also grown on borders of gardens, school compounds, office compounds, parks and residential sites. It could be established as Ornamental Avenue tree. The good capacity for pollarding and coppicing has made it suitable to fit in agro-forestry. The canopy of agar tree is such that it allows sunshine penetration partly. Thus, it can be planted in field boundaries, bunds etc., without affecting the field crops.



Agar in field boundary

Besides, agar tree has been successfully grown for strip planting along banks of ponds, tanks, canals and roads. The tree has been found suitable in horti-silvi-pastural system.

In hilly areas/tillas as in Barak valley it can be planted on poor soils on hill slopes, tilla tops. They help in reducing soil erosion and land sliding caused by rushing water during rainy season.

## AGAR IN TEA GARDENS

Tea growing situation is also ideal for agar tree. The increasing demand of agar tree as raw material, it is being introducing as shade tree in tea gardens particularly in Upper Assam with success. Agar



Agar as a shade tree in Tea Garden, Golaghat

tree is evergreen and with spreading canopy it allows partial penetration of sunshine through it. The lower branches are to be looped carefully at a regular intervals to avoid suppresion of tea bushes. The compatibility with the main crop however, has to be observed for a longer period. It has been observed the Agar tree has not exerted any ill-effects on the tea bushes up-to 10 th years. Agar can be planted in the borders and on both sides of internal roads of the gardens.

## PRESENT STATUS OF AGAR TREE IN THE N.E. INDIA

### Population Status and Trends

With the sudden shortfall in supply, price hike and increasing demand after 1970 in the oil rich Middle East, an indiscriminate exploitation of agar started throughout the species range. Only 10-15 % of the trees in any populations are likely to be infected with the fungus that cause the wood to undergo a change producing agar oil and/or agarwood. Traditionally in the past the local people have harvested only infected trees but demand in the last 20 year has led to excessive harvesting of both diseased and healthy trees.

This inflicted a heavy assault in the natural population, particularly in Assam and in the Hill forests of Khasi and Garo hills of Meghalaya, Nagaland, Manipur and Arunachal Pradesh, Tripura and Bhutan. Because of over exploitation and absence of professional approach of regeneration it has posed an alarming situation in its natural stand (wild population), particularly in Assam, the home of the finest quality agar.

According to the Indonesian National Forest Inventory populations are widespread in their range but patchy in distribution. The population densities of *Aquilaria spp.* are 1.87 individuals per hectare in Sumatra, 3.37 in Kalimantan and 4.33 in Irian Yaya. In Malaysia estimates lie at 2.5 individuals per hectare. (Soehartono in WCMC, 1997). In Indonesia the major centres of production are located at Riau and Aceh in Sumatra, also Kalimantan and Irian Yaya. The increasing rarity of the species has led to the traders searching in more remote areas even by helicopter and in some cases outside the species range. These resulted a steady transformation in its wild status (Table 1).

**Table 1** : Status of Agar tree in the wild

| Country/States | Status as on 24-5-94 |
|---|---|
| Bangladesh | Endangered |
| Bhutan | Rare |
| Myanmar | Vulnerable |
| Sumatra | Endangered |
| Singapore | Rare |
| Peninsular, Malaysia | Indeterminate |
| N.E.India : Assam | Rare |
| Manipur | Rare |
| Meghalaya | Rare |
| Tripura | Rare |

**Adapted from WCMC Status report as on 24-5-94**

In the Regional Seminar on Medicinal Plants, North-Eastern Region, Guwahati (Assam) held on 6th, 7th & 8th June, 1986 organized by "Union Ministry of Health and Family Welfare considering its fast depleting natural population a recommendation was passed on to the Govt. of India that the export of the products of the *A. agallocha* used in ISM & H , should be banned. The species *A. malaccensis* Lamk. has been included in the CITES list in Appendix II on 16th Feb. 1995. Based on these, the *Aquilaria spp.* have been placed under endangered category in India and imposed ban on *A.malaccensis,* Lam. for export by the Ministry of Commerce, Govt. of India vide notification no. 2(RE-98) 1997-2002 dated 13th April/1998. According to Asian Regional Workshop, 1997, the Indian populations are stated to be critically endangered. The following terms to specify natural population density (in wild) were adopted in the International Union for Conservation of Nature and Natural resources (IUCN) held in Stockholm and recorded in Red Data Book of Indian trees in the year 1972. IUCN Red List Categories of population density in wild are defined as follows:

**EXTINCT (EX):** A taxon is Extinct when there is no reasonable doubt that the last individual has died.

**EXTINCT IN THE WILD (EW):** A taxon is Extinct in the wild when it is known only to survive in cultivation, in captivity or as a naturalized population (or populations) well out side the past range. A taxon is presumed extinct in the wild when exhaustive surveys in known and/or expected habitat, at appropriate times (diurnal, seasonal, annual), throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.

**CRITICALLY ENDANGERED (CR):** A taxon is critically endangered when it is facing an extremely high risk of extinction in the wild in the immediate future.

***In Paradise Aloe-wood (Udul-Hind) will be burned as incense in the golden censers.***

***—AL-Bukhari***

**ENDANGERED (EN):** A taxon is endangered when it is not critically endangered but is facing a very high risk of extinction in the wild in the near future

**VULNERABLE (VU):** A taxon is Vulnerable when it is not critically Endangered or Endangered but is facing a high risk of extinction in the wild in the medium –term future.

**LOWER RISK (LR):** A taxon is lower risk when it has been evaluated, does not satisfy the criteria for any of the categories Critically endangered, Endangered or Vulnerable. Taxa included in the Lower Risk category can be separated into the subcategories:

(i) *Conservation dependent (cd):* Taxa which are the focus of a continuing taxon-specific or habitat-specific conservation programme targeted towards the taxon in question, the conservation of which would result in the taxon qualifying for one of the threatened categories above within a period of five years.

(ii) *Near Threatened (nt):* Taxa, which do not qualify for Conservation Dependent, but which are close to qualifying for Vulnerable.

(iii) *Least Concern (lc):* Taxa, which do not qualify for Conservation Dependent or near Threatened.

From the above definitions in strict sense it is seen that the *Aquilaria* species have wrongly placed under endangered category. They neither fall under Critically endangered/ endangered nor vulnerable nor threatened category. The species may be categorized under Lower Risk and Near Threatened (LR-nt). It is also not rare in its area of distribution. The natural regeneration through seeds is also quite fast (Troup, 1986)*. It has been exploited and over exploited but none of the situation mentioned above has been reached so far. In agar only the grown up trees are felled. Agar tree starts seeding very early right from 3rd year of planting. When a tree is felled for 'agaru' or agar oil extractions there are several saplings growing around it. Therefore, this population erosion from a range

* Troup, R.S., 1986. The Silviculture of Indian Trees, Vol. III Oxford University Press

area indicates that it is because of total deforestation and not because of felling agar trees only.

Authors enumerate the reasons of drastic reduction in wild population from a range area as under:

- Leasing of 'Agar Mahal' and allowing exploitation without proper monitoring which might have resulted deforestation atleast in some range areas.

- Absence of proper policy on the conservation and development of this valuable species.

- Doves are very much fond of its seeds. When the population of these birds was high natural regeneration was very less because of eating away maximum seeds by the birds. With the decrease in their population the natural regeneration rate is now increasing. This is one reason why the natural regeneration rate is higher in homestead or in agroforestry or plantation compared to wild.



'Natural regeneration of Agar tree'—A spot in Golaghat district showing the over crowded Agar tree *(Aquilaria agallocha)* growing naturally.

It has been experienced over ages that a species can be developed through its economic utilization on scientific basis only. Along with the decrease in its wild population it is increasing in many fold through private plantation.

## STATUS IN ARTIFICIAL REGENERATION

Agar tree has been emerged out as a high value economic species and with decreasing natural population commercial plantations are also coming up side by side. Now a days, it is sprung up as plantation crop and lakhs of saplings are being planted in the NE states every year since last two decades. Therefore, it is not likely that the species will come under any of the above categories shortly. A species can only be developed through its economic exploitation on scientific basis. Moreover, agar (sanchi) is such a plant that is deeply associated with the cultural heritage of this region. Even to-day one can find agar trees of 100 years old in homestead area particularly in Assam.

This shows that people of this region have realized the value of agar tree long back. At present, in NE India 100 per cent of raw material supply to the distillers and processors are coming from private plantations mostly from homestead planting and also through import. Of late, organized commercial plantation has paved the way. For conservation and development of these valuable species the following strategies may be undertaken:

i) To conduct survey on the availability of species, their variability and distribution and documentation.

ii) To introduce and promote as plantation crop (an immediate need) and encourage planting under private and corporate sectors as well.

iii) Collection of agar trees from natural population should be restricted only to the old and diseased trees.

iv) Collection and maintenance of germplasm at different agro-ecological situations should be done through the Forest Department. This will serve materials for future development and conserve against extinction if any.

v) R&D works should be initiated for further improvement in Govt. as well as in private sectors. To reduce the load on forest for its fire wood supply alternate fuel and fuel efficient distillation are to be worked out.

# PROMOTIONAL ACTIVITIES

Escalating demand and easy propagation of the species have tempted many growers to take up agar plantation on commercial basis in the entire North East region. In the natural stand its population has come down but at the same time its cultivation has increased many fold. Nagaland Govt. is promoting agar planting in the private sector on large scale through Corporation of Agar Plantation. Similar programme has also been undertaken by Govt. of Tripura through its Plantation Corporation. An NGO known as "DISHARI" located in South Tripura campaigning for replacing exotic rubber and teak programmes under Govt. patronization on commercial scale by the indigenous agar plantation from last few years. At present in Tripura starting from Assam border towards Dharmanagar many small to large size plantations in private holding are seen. There are some families owning more than 20,000 trees in their private plantation. In Barak valley the distillers mostly depend on the supply of quality raw material from these areas.

Markazul Ma-arif, a front ranking NGO based at Hojai under Nagaon district of Assam has been rendering multifarious social welfare activities is also doing a commendable role in popularizing agar as an organized plantation crop in Assam. The NGO has undertaken planting of agar tree in public places like parks, office and school compounds, holding group discussions, seminar under the technical guidance of the R & D Division as a part of their awareness campaign drive.

Besides experimental plantation at different locations the R&D division of the Flower Valley Agro-tech (P) Ltd of Ajmal Group, has successfully established plantation of different age at different sites totaling more than 70,000 trees by now. Mass scale commercial plantation under the Company is continuing. In a field survey it is also revealed that in upper Assam, a young tea planter Mr. Kartik Saikia of Kamarbandha in Golaghat district, has planted more than 1 lakh of agar saplings within last 7 years. He has raised agar as shade tree in tea, planted in garden boundaries, on both sides of

garden roads, as pure as well as mixed plantation.

Interestingly due to increase in population density particularly in Upper Assam the population of wood borer (*Zeuzera conferta*) followed by natural fungal infection is also increasing which have been confirmed by field studies. Now-a-days, villagers of many localities where there are abundance of Sanchi in homestad area earning a good income by selling infected trees of 10-15 years old to overcome their economic need. At present a large section of people in the region have started raising nurseries and plantations on commercial venture. Adoption of large scale plantation will also help in restoring eco-balance in the region. If this trend continues, it will change the present scenario of scarcity of agar wood to plenty for the industry.

## CULTURAL TREATMENT TO AUGMENT OIL FORMATION

Farmers have developed their own technique to make a very young tree harvestable. It is a common practice now a day, particularly in the traditional agar growing areas of upper Assam and also in Barak valley to apply mechanical injuries in the stem, branches at a regular interval for early infection. This is done just before breaking the dormancy i.e. before spring by giving a deep slanting cut with a sharp Dao or Axe. This is done to provide ready infection sites and also to push the tree to undergo a stress condition and in the following monsoon spreading of infection.

This practice yields better result where there is already built up microbial population in soil and also the climate is warm and humid. These cut injuries serve the initial sites of fungal infection. Generally, the 'Dum type' product obtained because of this treatment for oil extraction is locally known as "Ghap mal". This treatment shows better results in high rainfall areas. Because of high rainfall a favourable high humidity and temperature prevails which are conducive for early infection when a mechanical cut injury is provided. A 20 years old tree may produce only 5-10 kg of Dum'

without any treatment (which is due to natural injuries and mostly obtained at the junctures of broken branches) but when such treatment is applied the same tree is found to yield more than 30 kg after 2 years.

## DETECTION OF HARVESTABLE TREES

Since 'agaru' is located deep within the trunk, its detection from outer appearance is not easy. Generally, such trees are distinguished by some external features like poor and stunted crown development, the presence of swellings or cracks or depressions and cankers on the bole, signs of ill health particularly a die-back symptoms of the top and outer branches and a yellow tint to the woody tissues and also smaller leaves.



Deformed region in the stem- a site of agaru formation.

The visible wounds, stem distortions and the rotten branches provide evidences of 'agaru' deposits within a tree and finally the odour on examination by drawing samples with the help of screw augurs.

In deep alluvial soils or sandy loam soils the tree always shows

a luxuriant uninterrupted growth but in rocky soils, clay soils the tree looks undergoing certain amount of stress particularly during winter. These symptoms are very often associated with oil synthesis in the wood. There are no practical gadgets available so far for detecting the right tree to be felled. Only people who have developed the skill of identifying agar-bearing tree through experience detect trees for felling. Sometimes screw augers are driven inside at various depth and samples are drawn for examination.

## HARVESTING

Their physical age, growth rates and / or wood volume or physiological maturity does not govern the harvesting age of agar tree for commercial purpose. The whole tree does not yield agaru. Agar products of commercial interest are the products of disease caused by a group of fungi on agar plant. It is the infected tree and whose further growth is arrested due to physiological imbalance is harvested and for extracting agarwood and oil.

In a population all the tree do not get infected at a time and to the same extent. Under natural conditions infection starts from the age of 5 years (or even early) onwards. Agarwood formation takes longer time upto 30 to 40 years or even more. It is the advance stage of oil formation. More is the time the tree gets after infection more will be the yield of good quality agarwood. In order to get higher harvestable wood from each tree either agarwood or agar oil it is desirable to induce stress (nutrients /water/injuries) to the tree after 8 to 10 years of growth or even early. Before imposition of any amount of stress to the tree it is to be made more susceptible for fungal infection. This may be achieved by applying more nitrogenous fertilizers prior to the stress.

The younger tree even after infection yields essential oil containing higher amount of fatty oils/fatty acids which is undesirable from perfumery point of view. The oil quality goes on improving with time.

Generally, the bad and deformed trees attain harvestable first

unlike other forest species. The healthy trees are left to undergo stresses or subject to infection either naturally or artificially to induce oil formation. The harvesting is done on selection and continues for a longer period from a plantation raised at the same time.

## HARVESTING SEASON

Although the collection of agar trees for oil extraction as well as for 'agaru' is done almost throughout the year, the best time is during February-May, the dry season when the plant remain dormant or less active. During this period maximum concentration of oil with less waxy substances is obtained. The extracted oil during dry season possesses the finest odour and note compared to that obtained during rainy season when the plant remain active in growth.

## YIELD

The yield of commercial products of agar tree is not uniform per tree. It varies greatly and almost unpredictable. After 10 years of planting with intensive management each infected tree may yield about 30-40 kg 'Dum type' to Kolagachi' product for oil extraction, depending on infection intensity. In some areas the whole tree is processed for distillation. The *A. agallocha* in upper Assam yields better compared to other parts of the region. The oil contains higher proportion of fatty oils/acids when harvested early. Under present market price, from a 10 years crop cycle a net return of Rs. 17,45,535/ ha may be obtained. This gives a net return of Rs. 1,74,553/ and Rs. 5.26 per ha per year over Rupee investment, which is higher, compared to many other plantation crops in the state. From a 15-year crop cycle the net return over Rupee investment is still higher which comes about Rs. 7.59.

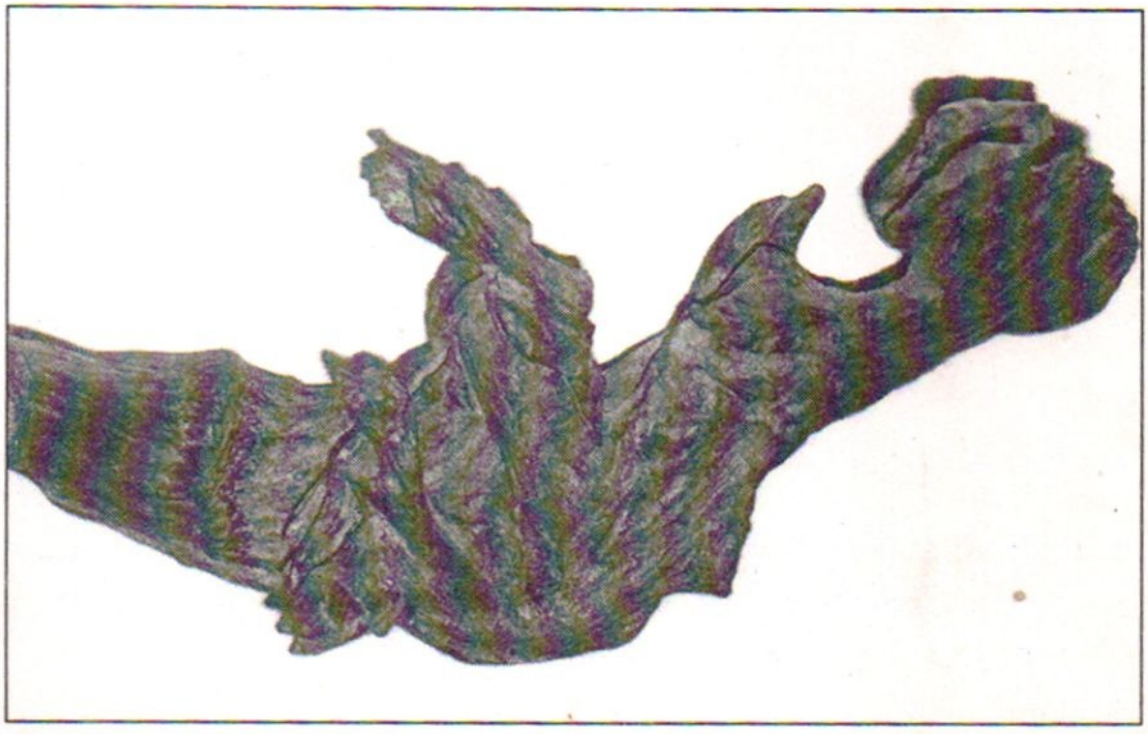
## FUTURE PROSPECTS AS PLANTATION CROP

The agro-climate and edaphic conditions of the state being ideal for plantation crops, the Agar plant or Sanchi plant would be adopted

by majority of the farmers as a remunerative cash crop. Once established as a regular plantation crop it will paved the way for excellent venture of useful employment generation particularly for the educated youths. Inclusion of the Sanchi plant as a regular plantation crop in the similar line of other plantation crops viz., Tea, Rubber, Coffee etc. will attract many to take up the plantation on commercial scale. A plantation of 10-15 years cycle could be an economically sound programme.

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***Sanchi the natural heritage of Assam.***
***Keep it up by planting more Sanchi plants***



Remains of a dead plant - a piece of highly charged 'Agaru' weighing 950 g.